Quantitative EEG and ERP report

CLINIC C LOGO

Patient:	Mock Patient 042
Date of birth:	1 Jan 2016
Age:	10 years
Sex:	female
Recorded:	15 Sep 2026
Case code:	MOCK-042
Earlier measurements:	23 Sep 2025 and 12 Mar 2026

CLINIC C LOGO

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This report presents the EEG, ERP and VCPT measurements of Mock Patient 042, aged 10, recorded on 15 Sep 2026.

Report	R-MOCK-042-0915 v1 · Issued	Recording system	Mitsar amplifier, WinEEG file (mockup label)
Template	CT-STD v3 · Clinic C · English	Result set	RS-MOCK-042-0915-1
Software	V4 mockup 0.1	Marking	Benjamini-Hochberg, q = 0.05 within each analysis block (stand-in)
Device identifier (UDI-DI)	UDI: placeholder, assigned at certification	Input checksum (sha256 of the de-identified signal)	`952da8c6638e5ea71e51566807d4fb613855c696f508314a3478ebba07739499` (synthetic input, not a recording file)
Pipeline	mockup pipeline 0.1.0	ICA seed	33226 (derived from the recording and stored)
Norms	Mock norms 2026.1 (synthetic); dataset version 2026.1	Processed	15 Sep 2026, 10:46, in 400 s
Text library	2026.1		
Device blocks	Device blocks 2026.1		

I. Summary

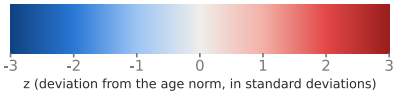
The largest marked deviations from the age norm in this measurement are: beta1 coherence, eyes closed: T3-T5 (z = 4.8), T5-P3 (z = 4.6), T5-O1 (z = 4.6), P3-O1 (z = 4.2), T3-C3 (z = 4.1), C3-T5 (z = 3.9), C3-P3 (z = 3.9); SMR asymmetry index, eyes open: C3-C4 (z = 4.2); beta1 coherence, eyes open: P3-O1 (z = 4.2), T3-C3 (z = 4.1), T5-O1 (z = 4.1), T3-T5 (z = 3.8), C3-T5 (z = 3.8), C3-P3 (z = 3.7), T5-P3 (z = 3.7); SMR asymmetry index, eyes closed: C3-C4 (z = 4.2); N1 amplitude, Go: O1 (z = -4.2), O2 (z = -4.2); beta2 coherence, eyes open: P4-O2 (z = 3.9); relative theta power, eyes open: P3 (z = -3.5), P4 (z = -3.4), O1 (z = -3.4), O2 (z = -3.3), Pz (z = -3.2).

This list ranks the marked values of these analyses by |z|: absolute and relative band power, maps per 1 Hz band, alpha peak frequency, theta/beta ratio, frontal alpha asymmetry index, asymmetry map, coherence per electrode pair, ERP peak amplitude and latency, and VCPT task results. It shows the 5 largest and every marked value of 3 z or more. Not ranked: the spectra per 0.5 Hz bin, shown as curves against the norm, and global coherence.

Marked values in this measurement: band power 5 of 456, maps per 1 Hz band 0 of 1,102, alpha peak and ratios 0 of 6, asymmetry map 2 of 96, coherence 15 of 2,052, ERP peaks 2 of 21, VCPT task 0 of 4, spectrum per 0.5 Hz bin 0 of 2,242 and global coherence 0 of 12.

Figure 1: Largest marked deviations, as in the summary

Analysis and condition	Electrode or pair, z
beta1 coherence, eyes closed	T3-T5 **4.8** T5-P3 **4.6** T5-O1 **4.6** P3-O1 **4.2** T3-C3 **4.1** C3-T5 **3.9** C3-P3 **3.9**
SMR asymmetry index, eyes open	C3-C4 **4.2**
beta1 coherence, eyes open	P3-O1 **4.2** T3-C3 **4.1** T5-O1 **4.1** T3-T5 **3.8** C3-T5 **3.8** C3-P3 **3.7** T5-P3 **3.7**
SMR asymmetry index, eyes closed	C3-C4 **4.2**
N1 amplitude, Go	O1 **-4.2** O2 **-4.2**
beta2 coherence, eyes open	P4-O2 **3.9**
relative theta power, eyes open	P3 **-3.5** P4 **-3.4** O1 **-3.4** O2 **-3.3** Pz **-3.2**



Reading key: z is the distance of a value from the normative mean in standard deviations of the normative sample. Cell and map colours show z on the fixed scale (blue below, red above the age norm). Bold with a ring: a value marked after correction for multiple comparisons. Dotted line in the maps: |z| = 2, which alone does not mark a value.

II. Signal quality and processing

Eyes open: 164 of 180 seconds were analysed after artefact removal.

Eyes closed: 171 of 180 seconds were analysed after artefact removal.

VCPT: 1168 of 1220 seconds were analysed after artefact removal.

Accepted VCPT trials: Go 95, NoGo 92, Ignore 93, Novelty 95 (of 100 each).

Artefact removal changed by the clinician: component 6 (muscle, left temporal) was kept although the automatic cleaning proposed removing it; 2 of 18 components were removed.

The analysed segments were confirmed by the clinician as proposed.

Alpha peak frequency, eyes closed (posterior mean): suggested 9.2 Hz, confirmed unchanged by the clinician; not marked (z = 0.0).

III. Anamnesis and clinical interview

Anamnesis	Synthetic example text. Seen for a quantitative EEG and ERP assessment at the family's request; third measurement at this clinic.
History	Synthetic example text. No relevant events reported by the parents.
Medication	Synthetic example text. None reported.
Current symptoms	Synthetic example text. Sleep and appetite as usual, as reported by the parents.

IV. Spectra and maps

Eyes open

Figure 2: Absolute power, z against the age norm, Eyes open

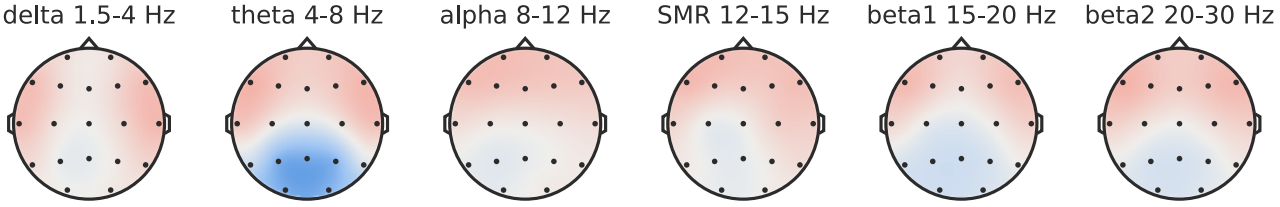
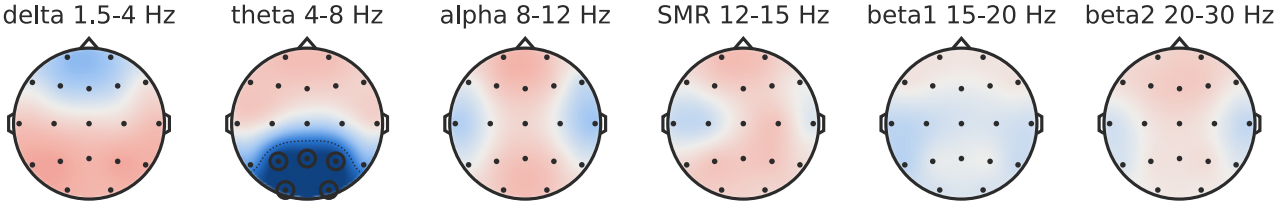


Figure 3: Relative power, z against the age norm, Eyes open



Relative theta power, eyes open, is below the age norm at P3 (z = -3.5), Pz (z = -3.2), P4 (z = -3.4), O1 (z = -3.4) and O2 (z = -3.3).

The alpha peak frequency, eyes open, is 9.1 Hz and is not marked (z = 0.0).

Eyes closed

Figure 4: Absolute power, z against the age norm, Eyes closed

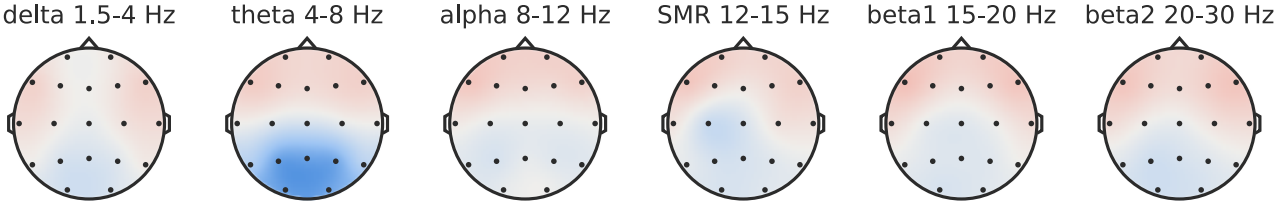
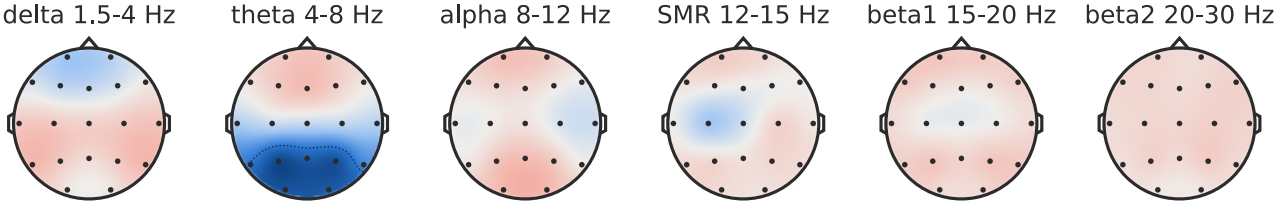
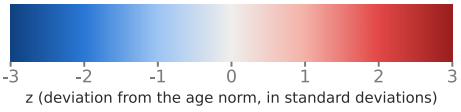


Figure 5: Relative power, z against the age norm, Eyes closed



No absolute or relative band power value, eyes closed, is marked.

The alpha peak frequency, eyes closed, is 9.2 Hz and is not marked (z = 0.0).

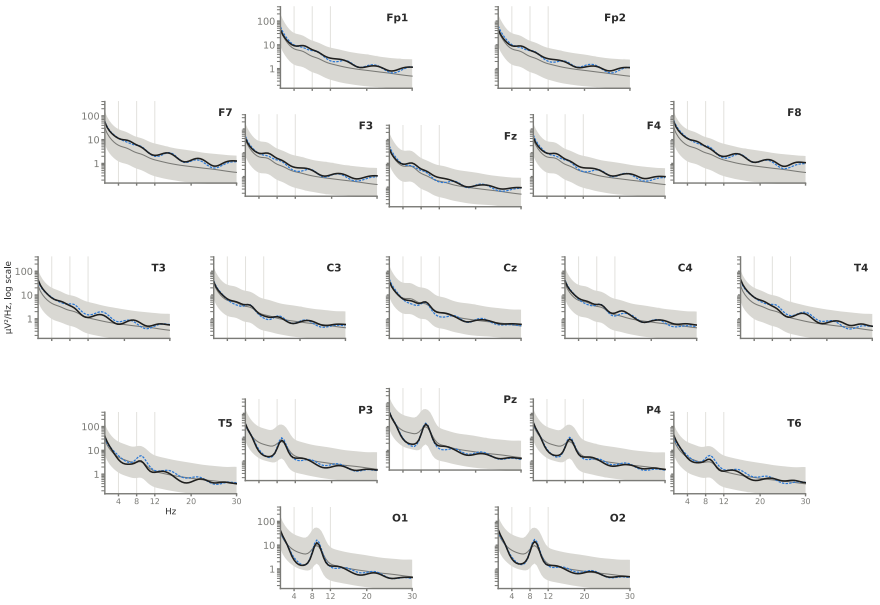


Ring: marked value (after correction for multiple comparisons).

V. Absolute spectra per electrode

Eyes open

Figure 6: Absolute spectra per electrode, Eyes open



Eyes closed

Figure 7: Absolute spectra per electrode, Eyes closed

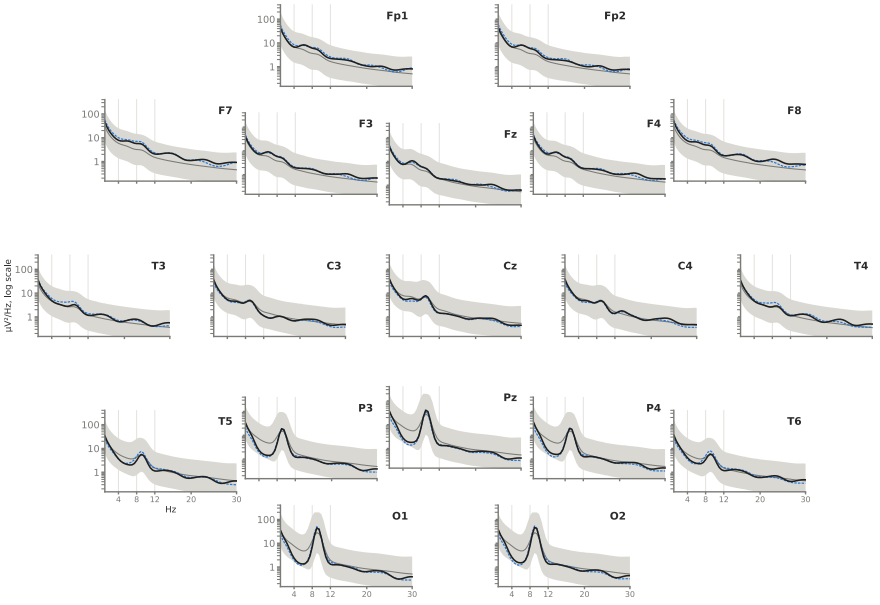


Figure 8: Relative spectra per electrode (share of total power per 0.5 Hz bin), Eyes open

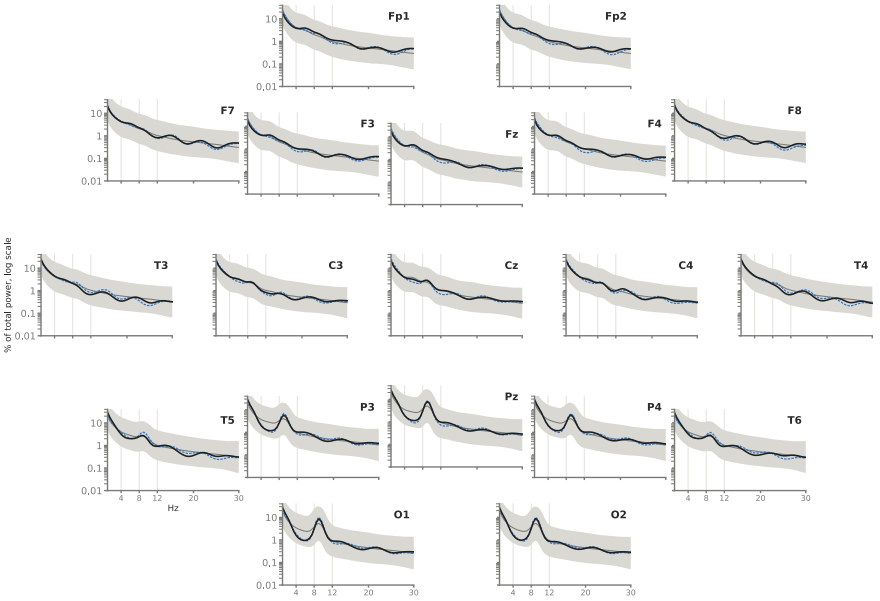
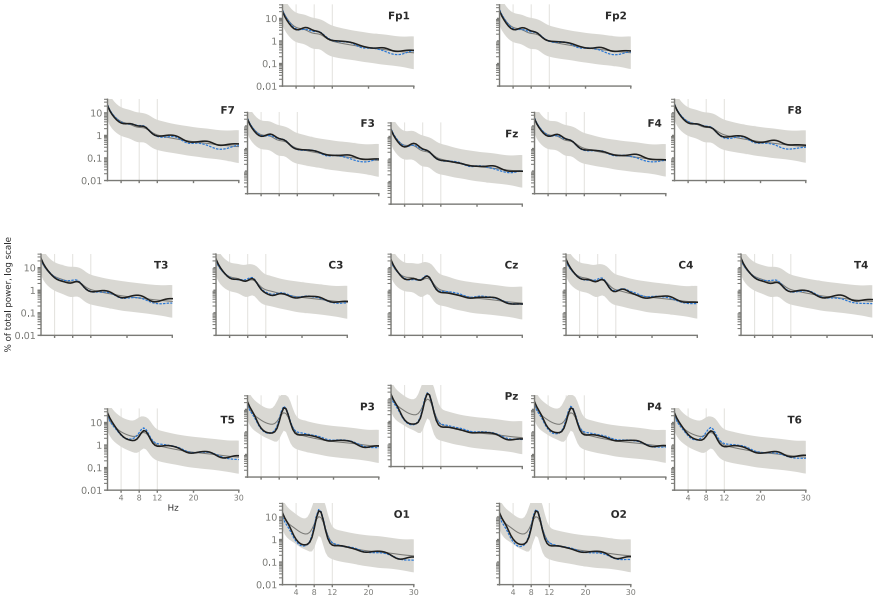


Figure 9: Relative spectra per electrode (share of total power per 0.5 Hz bin), Eyes closed



Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Blue dashed: earlier measurement. Norm band of the relative spectra and z of the relative maps per 1 Hz band: a stand-in derived from the absolute norm in this mockup.

VI. Frequency markers

Figure 10: Frequency markers

No.	Condition	Electrode	Frequency (Hz)	z	Question, answer	Choice
1	Eyes open	C4	13.5	0.7	-	suggested, confirmed
2	Eyes open	O2	9.0	0.5	-	suggested, confirmed
3	Eyes open	C4	21.5	0.4	-	suggested, confirmed
1	Eyes closed	F3	6.0	0.6	-	suggested, confirmed
2	Eyes closed	O2	9.0	0.5	-	suggested, confirmed
3	Eyes closed	C4	23.0	0.4	-	suggested, confirmed

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 11: z at the marker frequency, Eyes open

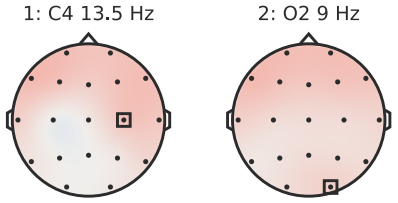
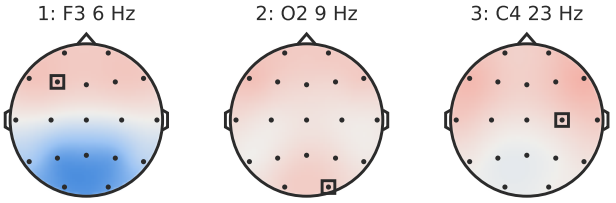


Figure 12: z at the marker frequency, Eyes closed



Square: the electrode of the marker. Ring: marked value (after correction for multiple comparisons).

Frequency markers, eyes open: suggested by the system and confirmed unchanged by the clinician; markers: 3.

Frequency markers, eyes closed: suggested by the system and confirmed unchanged by the clinician; markers: 3.

The system proposes up to 3 peaks of the spectrum per condition, those with the largest positive z (absolute power per 0.5 Hz bin); the clinician confirms, moves, removes or adds markers. Each marker shows its z and whether that value is marked.

VII. Maps per 1 Hz band (absolute power, z)

Figure 13: Maps per 1 Hz band (absolute power, z), Eyes open

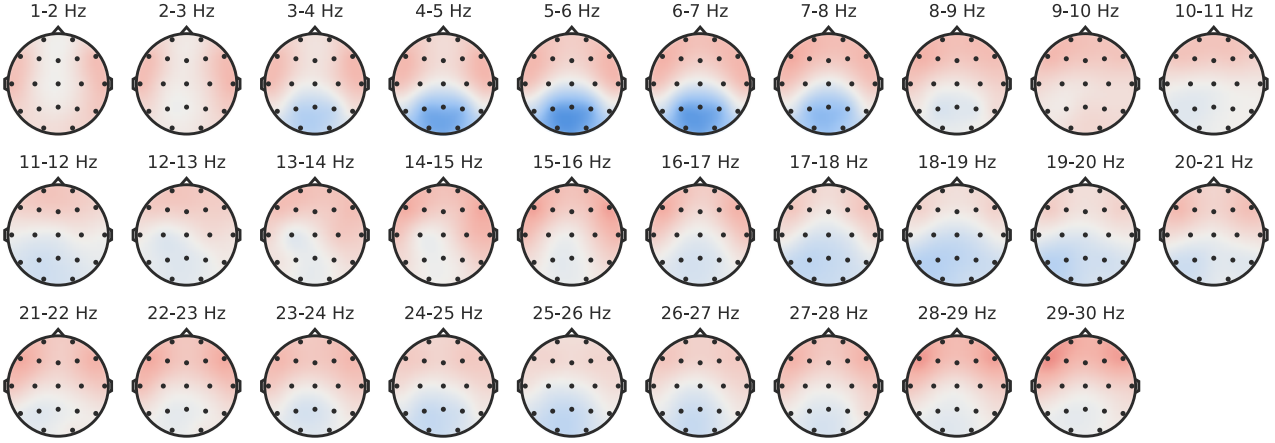
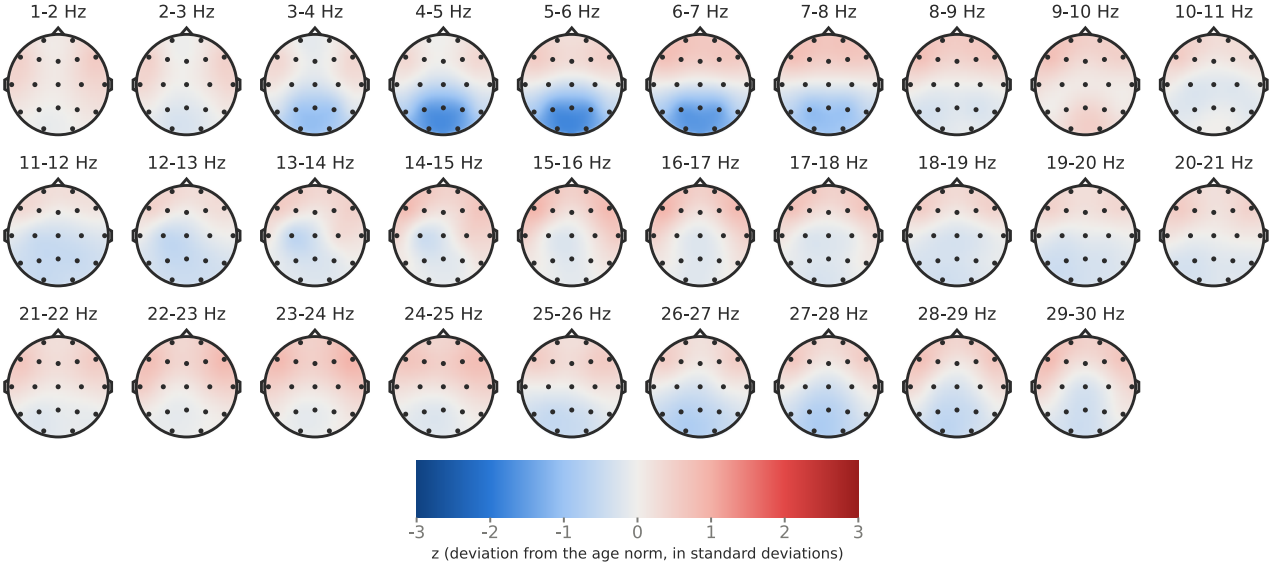


Figure 14: Maps per 1 Hz band (absolute power, z), Eyes closed



Ring: marked value (after correction for multiple comparisons).

VIII. Band ratios and indices

Measure		Value	Norm mean	z
Theta/beta ratio at Cz	Eyes open	2.54	2.76	-0.3
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes open	-0.06	-0.01	-0.6
Alpha peak frequency (Hz)	Eyes open	9.1	9.1	0.0
Theta/beta ratio at Cz	Eyes closed	2.66	2.82	-0.2
Frontal alpha asymmetry index (ln F4 minus ln F3)	Eyes closed	-0.07	-0.01	-0.6
Alpha peak frequency (Hz)	Eyes closed	9.2	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

The theta/beta ratio at Cz, eyes open, is 2.54 (z = -0.3).

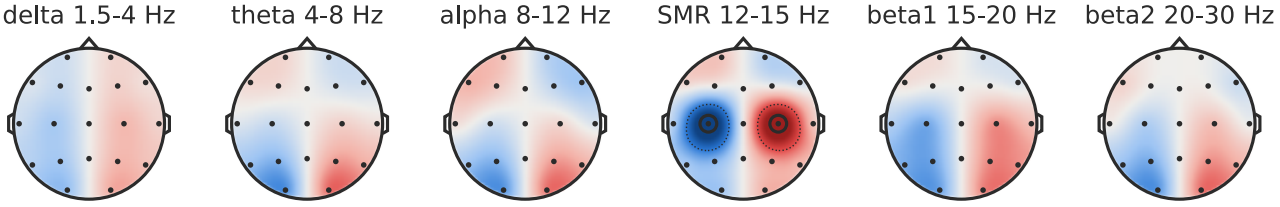
The frontal alpha asymmetry index (ln F4 minus ln F3), eyes open, is -0.06 (z = -0.6).

The theta/beta ratio at Cz, eyes closed, is 2.66 (z = -0.2).

The frontal alpha asymmetry index (ln F4 minus ln F3), eyes closed, is -0.07 (z = -0.6).

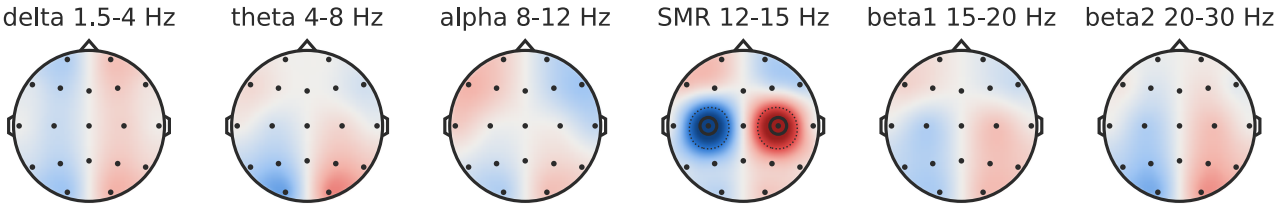
IX. Asymmetry

Eyes open



The SMR asymmetry index C3-C4, eyes open, is 0.24 (more power at C4), above the age norm (z = 4.2).

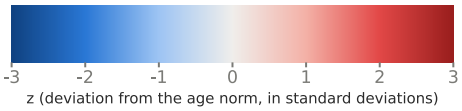
Eyes closed



The SMR asymmetry index C3-C4, eyes closed, is 0.25 (more power at C4), above the age norm (z = 4.2).

Figure 15: Marked electrode pairs

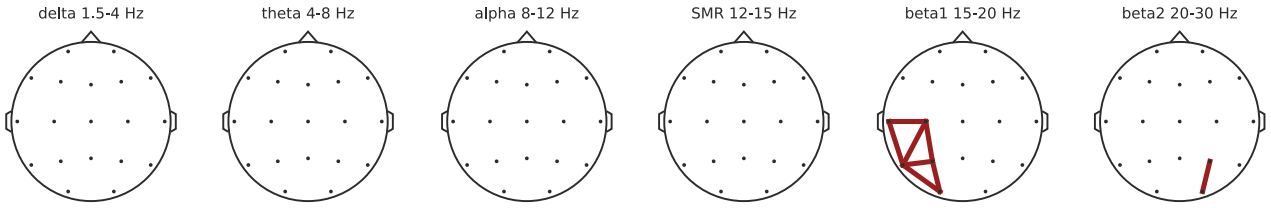
Condition	Band	Pair	Index	More power at	z
Eyes open	SMR	C3-C4	0.24	C4	○ **4.2**
Eyes closed	SMR	C3-C4	0.25	C4	○ **4.2**



Asymmetry index (right minus left) / (right plus left) per electrode pair: its z against the age norm at the right electrode of the pair and the same z with the opposite sign at the left electrode; midline 0. Ring: marked value (after correction for multiple comparisons).

X. Coherence

Eyes open



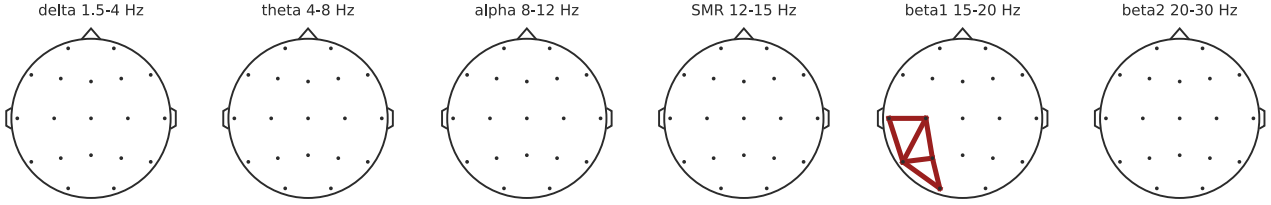
Beta1 coherence, eyes open, is above the age norm for 7 electrode pairs: P3-O1 (z = 4.2), T3-C3 (z = 4.1), T5-O1 (z = 4.1), T3-T5 (z = 3.8), C3-T5 (z = 3.8), C3-P3 (z = 3.7) and 1 more in the table.

Beta2 coherence, eyes open, is above the age norm for 1 electrode pair: P4-O2 (z = 3.9).

Figure 16: Marked electrode pairs, Eyes open

Band	beta1							beta2
Pair	P3-O1	T3-C3	T5-O1	T3-T5	C3-T5	C3-P3	T5-P3	P4-O2
Coherence	0.73	0.69	0.64	0.62	0.60	0.64	0.70	0.68
z	○ **4.2**	○ **4.1**	○ **4.1**	○ **3.8**	○ **3.8**	○ **3.7**	○ **3.7**	○ **3.9**

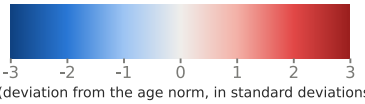
Eyes closed



Beta1 coherence, eyes closed, is above the age norm for 7 electrode pairs: T3-T5 (z = 4.8), T5-P3 (z = 4.6), T5-O1 (z = 4.6), P3-O1 (z = 4.2), T3-C3 (z = 4.1), C3-T5 (z = 3.9) and 1 more in the table.

Figure 17: Marked electrode pairs, Eyes closed

Band	beta1						
Pair	T3-T5	T5-P3	T5-O1	P3-O1	T3-C3	C3-T5	C3-P3
Coherence	0.68	0.76	0.67	0.73	0.69	0.61	0.66
z	○ **4.8**	○ **4.6**	○ **4.6**	○ **4.2**	○ **4.1**	○ **3.9**	○ **3.9**



Lines: marked electrode pairs, coloured by z. Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

XI. Event-related potentials at the electrodes (channel ERPs)

Figure 18: Channel ERPs, Go trials

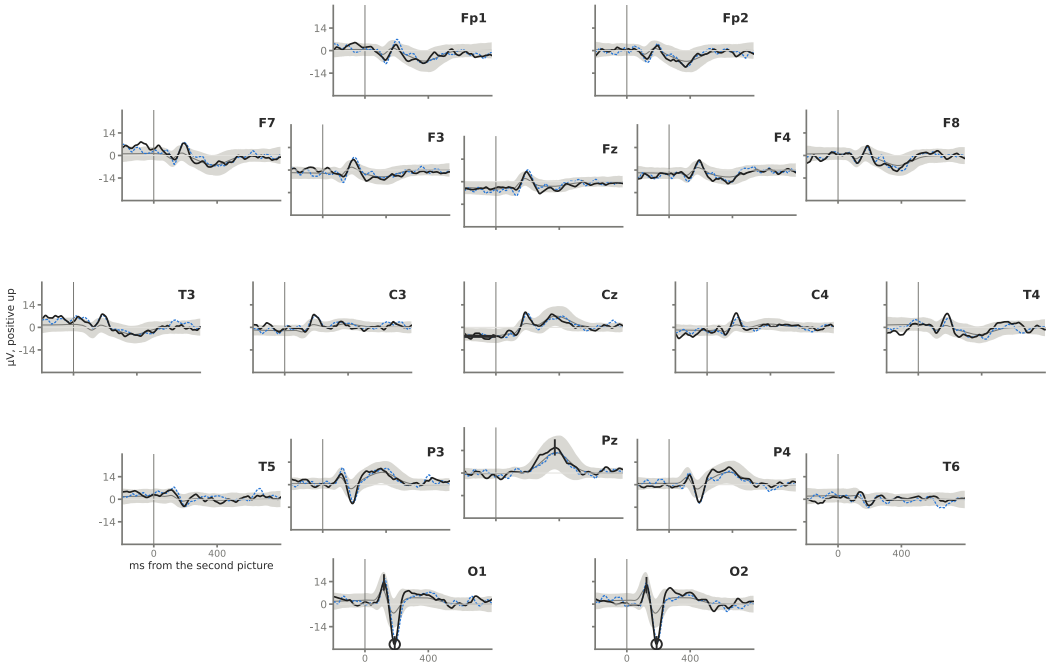


Figure 19: Channel ERPs, NoGo trials

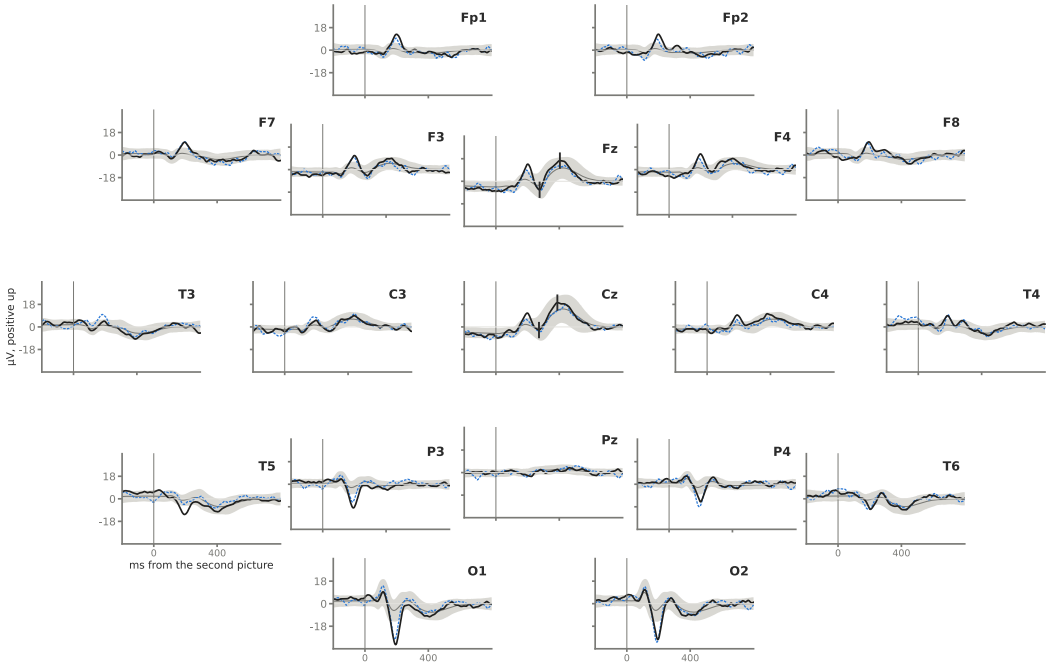


Figure 20: Channel ERPs, Novelty trials

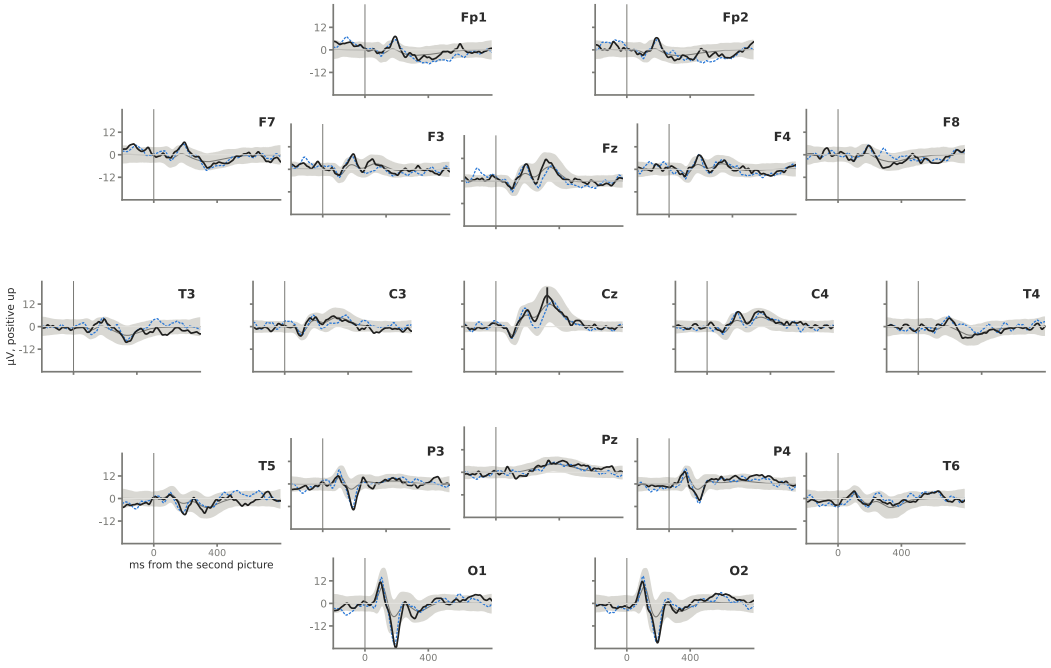


Figure 21: Go and NoGo trials and the difference Go minus NoGo (no norm)

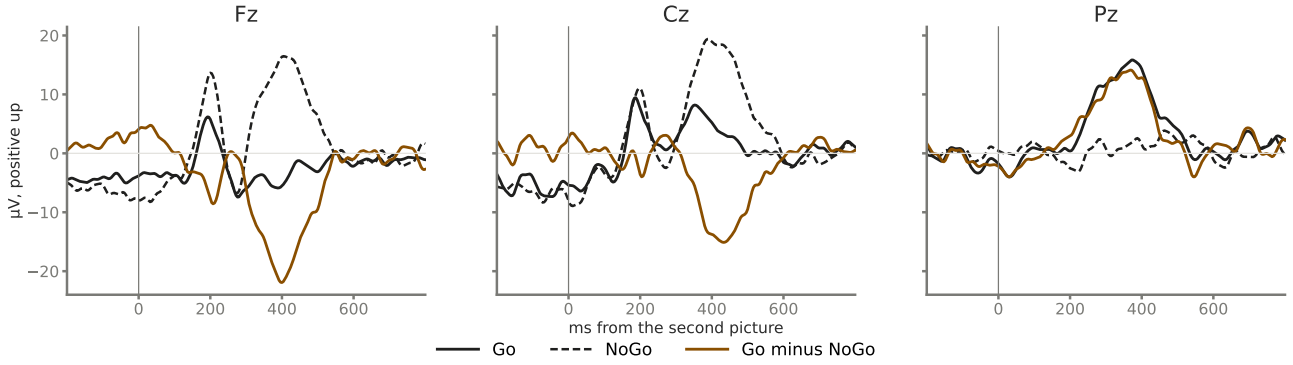


Figure 22: ERP peak values

Component	Trial type	Electrode	Amplitude (µV)	z	Latency (ms)	z
P1	Go	O1	13.5	0.2	120	0.5
P1	Go	O2	11.8	-0.2	124	0.7
N1	Go	O1	-25.0	○ **-4.2**	188	0.3
N1	Go	O2	-24.9	○ **-4.2**	188	0.3
N2	NoGo	Fz	-6.8	0.0	276	0.3
N2	NoGo	Cz	-2.5	0.3	272	0.3
P3	Go	Pz	15.9	0.1	372	-0.1
P3	NoGo	Cz	19.4	0.3	388	-0.7
P3	NoGo	Fz	16.5	0.8	404	-0.3
P3a	Novelty	Cz	16.6	0.4	324	-0.2
CNV	Go	Cz	-5.6	-0.5		

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

For N1, N2 and CNV a negative z means a larger negativity. CNV: mean amplitude from -200 to 0 ms before the second picture; no latency.

Black: this measurement. Grey line and band: normative mean and mean ± 2 SD. Time 0: second picture. Tick: scored peak; ring: marked value; bar: CNV window at its mean. Blue dashed: earlier measurement.

ERP peaks, Go trials: the suggested peaks were confirmed unchanged by the clinician.

ERP peaks, NoGo trials: the suggested peaks were confirmed unchanged by the clinician.

ERP peaks, Novelty trials: the suggested peaks were confirmed unchanged by the clinician.

N1 amplitude at O1, Go trials, is more negative than the age norm (-25.0 µV, z = -4.2).

N1 amplitude at O2, Go trials, is more negative than the age norm (-24.9 µV, z = -4.2).

XII. Checklist over channel ERPs

Component	Trial type	Electrode	Window (ms)	Reviewed	Peak identifiable	Latency inside the norm band	Comment
P1	Go	O1	70 to 170	yes	yes	yes	
P1	Go	O2	70 to 170	yes	yes	yes	
N1	Go	O1	140 to 250	yes	yes	yes	
N1	Go	O2	140 to 250	yes	yes	yes	
N2	NoGo	Fz	210 to 330	yes	yes	yes	
N2	NoGo	Cz	210 to 330	yes	yes	yes	
P3	Go	Pz	280 to 560	yes	yes	yes	
P3	NoGo	Cz	300 to 600	yes	yes	yes	
P3	NoGo	Fz	300 to 600	yes	yes	yes	
P3a	Novelty	Cz	250 to 420	yes	yes	yes	
CNV	Go	Cz	-200 to 0	yes	yes	-	

Entries by the clinician, against the norm curves of the channel ERP figures. CNV has no latency. Comments are the clinician's own text.

Checklist over channel ERPs: completed by the clinician; rows reviewed: 11 of 11.

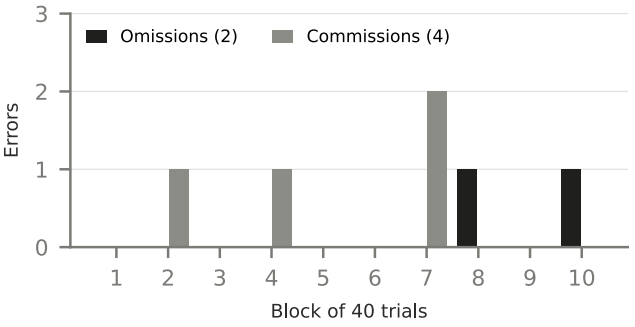
In this checklist the clinician records for each scored channel ERP whether it was reviewed, whether its peak can be identified and whether its latency lies inside the norm band; the system computes nothing from the entries and states them as entered.

XIII. VCPT task performance

Measure	Value	Percentile	z
Omission errors	2	46	-0.1
Commission errors	4	81	0.9
Mean reaction time	409 ms	47	-0.1
Reaction time variability (SD)	86 ms	42	-0.2

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 23: Errors per block of 40 trials (no norm)



Omission errors: 2, not marked (percentile 46).

Commission errors: 4, not marked (percentile 81).

Mean reaction time: 409 ms, not marked (percentile 47).

Reaction time variability (SD): 86 ms, not marked (percentile 42).

XIV. Comparison of measurements

Measure	Earlier 23 Sep 2025 (uploaded later)	z	Earlier 12 Mar 2026	z	Current 15 Sep 2026	z
Omission errors	3	0.3	3	0.3	2	-0.1
Commission errors	6	1.5	5	1.2	4	0.9
Mean reaction time	431 ms	0.2	421 ms	0.1	409 ms	-0.1
Reaction time variability (SD)	97 ms	0.2	95 ms	0.2	86 ms	-0.2
Theta/beta ratio at Cz, Eyes open	2.54	-0.3	2.22	-0.7	2.54	-0.3
Alpha peak frequency (Hz), Eyes open	9.1	0.1	9.1	0.0	9.1	0.0
Theta/beta ratio at Cz, Eyes closed	2.47	-0.4	2.26	-0.7	2.66	-0.2
Alpha peak frequency (Hz), Eyes closed	9.3	0.3	9.2	0.0	9.2	0.0

Bold with a ring: marked z value (after correction for multiple comparisons); the cell colour shows z.

Figure 24: Earlier 23 Sep 2025: Relative power, z against the age norm, Eyes open

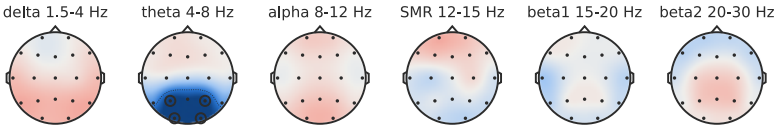


Figure 25: Earlier 12 Mar 2026: Relative power, z against the age norm, Eyes open

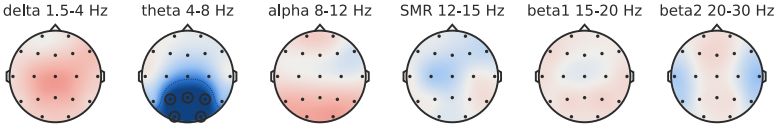
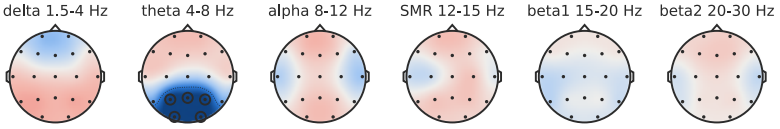


Figure 26: Current 15 Sep 2026: Relative power, z against the age norm, Eyes open



Ring: marked value (after correction for multiple comparisons).

Beta1 coherence, eyes closed, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: C3-P3 4.5, 5.0 and 3.9; T3-T5 4.4, 4.7 and 4.8; C3-T5 4.5, 4.8 and 3.9; T3-C3 4.5, 4.7 and 4.1; T5-P3 4.1, 4.5 and 4.6; T5-O1 4.4, 4.6 and 4.6; P3-O1 4.3, 4.4 and 4.2.

Beta1 coherence, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: T3-C3 4.1, 4.8 and 4.1; C3-T5 4.8, 4.3 and 3.8; T3-T5 4.2, 4.5 and 3.8; T5-P3 4.3, 4.5 and 3.7; T5-O1 4.3, 4.3 and 4.1; C3-P3 4.3, 4.3 and 3.7; P3-O1 4.2, 4.2 and 4.2.

N1 amplitude, Go, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: O2 -4.4, -3.9 and -4.2; O1 -4.2, -3.7 and -4.2.

SMR asymmetry index, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: C3-C4 3.9, 3.6 and 4.2.

SMR asymmetry index, eyes closed, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: C3-C4 3.9, 3.6 and 4.2.

Beta2 coherence, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: P4-O2 0.7, 0.5 and 3.9.

Relative theta power, eyes open, z on 23 Sep 2025, 12 Mar 2026 and 15 Sep 2026: P4 -3.3, -3.7 and -3.4; P3 -3.4, -3.6 and -3.5; Pz -3.0, -3.6 and -3.2; O1 -3.4, -3.3 and -3.4; O2 -3.3, -3.3 and -3.3.

This comparison lists 24 of the 24 values marked in at least one of the measurements, ranked by |z|: the 6 largest and every value marked at 3 z or more.

The earlier measurement of 23 Sep 2025, uploaded later, was processed with the current software and norms (V4 mockup 0.1, mockup pipeline 0.1.0; Mock norms 2026.1 (synthetic)) as result set RS-MOCK-042-0923-1.

The earlier measurement of 12 Mar 2026 was processed again with the current software and norms (V4 mockup 0.1, mockup pipeline 0.1.0; Mock norms 2026.1 (synthetic)) as result set RS-MOCK-042-0312-2; result set RS-MOCK-042-0312-1, from which report R-MOCK-042-0312 v1 was issued, stays unchanged.

Measurements are shown side by side, each against its own age norm; differences between them are not tested for statistical significance.

Measurements uploaded later are processed from their recordings; reports and numbers made by earlier software are not imported.

XV. Notes on the results

The results are one input to the clinician's assessment. The software does not diagnose and does not recommend treatment.

z is the distance of a value from the normative mean, in standard deviations of the normative sample; about 95% of the normative sample lies between z = -2 and z = 2.

Marked values differ from the age norm after correction for multiple comparisons, which keeps chance results among the many comparisons rare; a value beyond 2 z that is not marked may deviate by chance alone. The method of correction is still to be chosen.

XVI. Method and norms

Recording: Mitsar amplifier, WinEEG file (mockup label); 19 scalp electrodes of the 10-20 system (Fp1 to O2); conditions: eyes open, eyes closed and VCPT.

Processing: average reference of the 19 scalp channels; band-pass filter 0.5 to 45 Hz and mains notch filter at 50 Hz; analysis rate 250 Hz; artefacts removed by independent component analysis (ICA) with reference maps, then remaining artefact segments excluded.

Spectra: 1 to 30 Hz in steps of 0.5 Hz from windows of 2 s with 50 % overlap, compared with the norm as log10 power; relative power is the share of the total power from 1.5 to 30 Hz. Bands: delta 1.5 to 4 Hz, theta 4 to 8 Hz, alpha 8 to 12 Hz, SMR 12 to 15 Hz, beta1 15 to 20 Hz and beta2 20 to 30 Hz. Theta/beta ratio at Cz with theta 4 to 8 Hz and beta 13 to 21 Hz.

ERPs: VCPT with 100 trials each of Go, NoGo, Ignore and Novelty; averages of correct trials only; epochs from -200 to 800 ms around the second picture, baseline before the cue; peaks read off the curves in fixed windows (P1 70 to 170 ms, N1 140 to 250 ms, N2 210 to 330 ms, P3 (Go) 280 to 560 ms, P3 (NoGo) 300 to 600 ms, P3a 250 to 420 ms); CNV as the mean from -200 to 0 ms.

Norms: Mock norms, version 2026.1 (synthetic); age range of the norms 6 to 67 years; norm group: 400 synthetic subjects per condition at the age at recording, 10 years.

Marking: a value is marked when it differs from the age norm after correction for multiple comparisons; until the method is chosen, this mockup uses a stand-in correction, the Benjamini-Hochberg false discovery rate (q = 0.05) within each analysis block.

Recording conditions (documented at upload)

Time of recording	09:40
Medication on the recording day	none reported
Time of the last dose	not applicable
Sleep the night before	as usual
Corrected vision (glasses or lenses)	not documented
Response hand (VCPT)	not documented
Technician's note: drowsiness during the recording	no

Issued by

Issued by	Clinician 1, licensed psychologist
Date of issue	18 Sep 2026, 14:20
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Version	1 · Issued
Replaces	none, first version

Text sources

System sentences in this report come from the fixed sentence library 2026.1. Sentence ids by section:

Summary `SUM-2, SUM-RULE-2, SUM-COUNT-1`

Signal quality and processing `Q-DUR-1, Q-TRIALS-1, Q-ICA-MAN-1, Q-SEG-AUTO-1, MK-AUTO-1`

Spectra and maps `SP-REL-1, APF-IN-2, SP-NONE-2`

Frequency markers `FM-AUTO-1, FM-RULE-1`

Band ratios and indices `TBR-1, FAA-1`

Asymmetry `ASY-1`

Coherence `COH-1`

Event-related potentials at the electrodes (channel ERPs) `ERP-PEAK-AUTO-1, ERP-AMP-NEG-1`

Checklist over channel ERPs `ERPCL-ROWS-1, ERPCL-RULE-1`

VCPT task performance `BEH-IN-2`

Comparison of measurements `CMP-G-1, CMP-RULE-1, L-REPROC-2, L-REPROC-1, L-SIDE-1, L-IMPORT-1`

Notes on the results `L-AID-1, L-Z-1, L-MC-1`

Method and norms `MET-REC-1, MET-PROC-1, MET-SPEC-1, MET-ERP-1, MET-NORM-1, MET-MC-1`